

Deploying via Workbench

The Deploy File

```
deploy/VerificationCodeLoginFlow_Deploy.zip
```

API Version: 62.0 | **Size:** ~18 KB | **Components:** 19

What's Inside the ZIP

```
VerificationCodeLoginFlow_Deploy.zip
├── package.xml                    ← manifest (API v62.0)
├── classes/
│   ├── VerificationCodeService.cls    ← core generate + validate logic
│   ├── VerificationCodeService.cls-meta.xml
│   ├── VerificationCodeGenerator.cls  ← flow wrapper (generate)
│   ├── VerificationCodeGenerator.cls-meta.xml
│   ├── VerificationCodeValidator.cls  ← flow wrapper (validate)
│   ├── VerificationCodeValidator.cls-meta.xml
│   ├── VerificationCodeChecker.cls    ← blocked-state check at login start
│   ├── VerificationCodeChecker.cls-meta.xml
│   ├── VerificationCodeCleanup.cls    ← scheduled daily cleanup
│   ├── VerificationCodeCleanup.cls-meta.xml
│   ├── VerificationCodeServiceTest.cls ← test class (13 tests)
│   ├── VerificationCodeServiceTest.cls-meta.xml
│   ├── VerificationCodeCheckerTest.cls ← test class (4 tests)
│   ├── VerificationCodeCheckerTest.cls-meta.xml
│   ├── VerificationCodeCleanupTest.cls ← test class (2 tests)
│   └── VerificationCodeCleanupTest.cls-meta.xml
├── objects/
│   └── Verification_Code__c.object    ← custom object + all 5 fields
├── flows/
│   └── VerificationCodeLoginFlow.flow ← login screen flow (Active)
```

Step-by-Step Workbench Deployment

Step 1 — Log in to Workbench

1. Go to <https://workbench.developerforce.com>
2. Set **Environment** to match your target org type (**Production** / **Sandbox**)

- 3. Set **API Version** to 62.0
- 4. Tick **I agree to the terms of service**
- 5. Click **Login with Salesforce** and authorise with an admin account

Step 2 — Navigate to Deploy

Migration → Deploy

Step 3 — Upload the ZIP

- 1. Click **Choose File** → select deploy/VerificationCodeLoginFlow_Deploy.zip
- 2. Set options:

Option	Value
Rollback On Error	☒ Checked
Single Package	☒ Checked
Allow Missing Files	☐ Unchecked
Run Tests	RunSpecifiedTests
Test Classes	VerificationCodeServiceTest, VerificationCodeCheckerTest, VerificationCodeCleanupTest
Purge On Delete	☐ Unchecked

- 3. Click **Next**

Step 4 — Confirm and Deploy

- 1. Review the component list — you should see **19 components**
- 2. Click **Deploy**
- 3. Wait 1–2 minutes
- 4. Confirm **Status: Succeeded** and **Number Components Deployed: 19** ☒

⚠ If you see **Number Components Deployed: 0** — the API version in Workbench doesn't match the ZIP. Make sure Workbench is set to **API 62.0** before uploading.

Step 5 — Manual Configuration (Required After Every Deploy)

The flow deploys as **Active** automatically, but the Login Flow assignment must be done manually.

5a — Assign to Standard Login

1. **Setup** → **Identity** → **Login Flows** → **New**
2. Flow: **Verification Code Login Flow**
3. Login Page: **Default Login Page** (or your My Domain)
4. Click **Save**

5b — Assign to SSO (Google SAML)

1. **Setup** → **Single Sign-On Settings**
2. Click **Edit** on your Google SAML configuration
3. Under **Login Flow**, select **Verification Code Login Flow**
4. Click **Save**

5c — Schedule the Cleanup Job

Developer Console → **Debug** → **Open Execute Anonymous Window** → paste and run:

```
System.schedule(
    'VerificationCodeCleanup Daily',
    '0 0 2 * * ?',
    new VerificationCodeCleanup()
);
```

Verify: **Setup** → **Scheduled Jobs** — **VerificationCodeCleanup Daily** should appear.

Regenerating the ZIP After Future Changes

⚠ Do NOT use PowerShell's **Compress-Archive** — it writes backslash paths into the ZIP which Workbench cannot parse. Use Python instead (Python 3 is required).

```
# 1. Convert SFDX source to Metadata API format
sf project convert source --source-dir force-app/main/default --output-dir
deploy/metadata

# 2. Replace with explicit package.xml
Copy-Item manifest/package.xml deploy/metadata/package.xml -Force

# 3. Remove unused components
Remove-Item -Recurse -Force deploy/metadata/lwc -ErrorAction SilentlyContinue
Remove-Item -Force deploy/metadata/classes/VerificationCodeBlocker.cls -
ErrorAction SilentlyContinue
Remove-Item -Force deploy/metadata/classes/VerificationCodeBlocker.cls-meta.xml -
ErrorAction SilentlyContinue

# 4. Rebuild ZIP using Python (ensures forward-slash paths for Workbench
compatibility)
python -c "
import zipfile, os
src = r'deploy\metadata'
```

```
dst = r'deploy\VerificationCodeLoginFlow_Deploy.zip'
with zipfile.ZipFile(dst, 'w', zipfile.ZIP_DEFLATED) as zf:
    for root, dirs, files in os.walk(src):
        for f in files:
            fp = os.path.join(root, f)
            arcname = os.path.relpath(fp, src).replace(os.sep, '/')
            zf.write(fp, arcname)
print('ZIP rebuilt successfully')
"
```

Troubleshooting

Error / Symptom	Fix
0 Components Deployed / 0 Total	Two possible causes: (1) Workbench API version doesn't match — set it to 62.0 . (2) ZIP was built with PowerShell Compress-Archive (backslash paths) — rebuild using the Python script above.
This schedulable class has jobs pending	Abort the existing VerificationCodeCleanup scheduled job then redeploy
Flow not visible in Login Flow dropdown	Flow must be Active . Go to Setup → Flows → Activate it.
field integrity exception: User__c	Field already exists with wrong constraints. Delete it in Setup → Object Manager and redeploy.
Required field is missing: sharingModel	The Verification_Code__c object failed to deploy. Check object deployment in Workbench results.